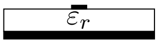
HF

1 Grundlagen

Mikrostreifenleitungen: Quasi-TEM-Wellen

Luft



Masse

Triplate Leitung: TEM-Wellen

Masse

Leitungsgleichung γ, Z_L

$U = U_f e^{-\gamma z} + U_b e^{\gamma z}$

$I = \frac{U_f}{Z_L} e^{-\gamma z} - \frac{U_f}{Z_L} e^{\gamma z}$

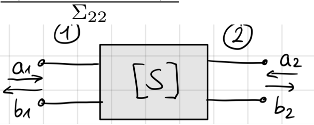
Normierungs:

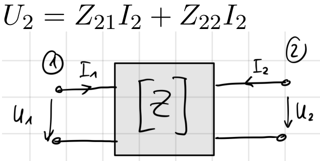
$u = \frac{U}{\sqrt{Z_L}}$, $i = I\sqrt{Z_0}$, $U_f = I_f Z_L$

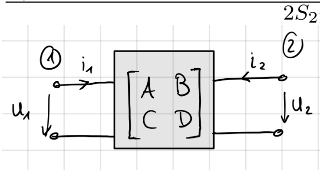
$u = u_f e^{-\gamma z} + u_b e^{\gamma z} = a + b$

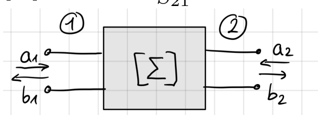
$i = a - b$

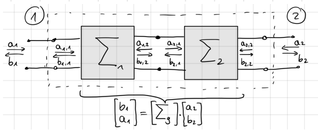
Matrixbeschreibungen

- **S-Parameter (Streuparameter)**:
  - Beschreibung über Wellen (a_i (Eingang Oben), b_i (Ausgang unter), Nummer sind die Seite)
  - $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$
    * $S_{11}\big|_{a_2=0} = \frac{b_1}{a_1}$: Reflexion Eingang
    * $S_{21}\big|_{a_2=0} = \frac{b_2}{a_1}$: Transmission (Verstärkung)
    * S_{12}: Rückwirkung
    * S_{22}: Reflexion Ausgang
  - $\frac{\begin{pmatrix} A+B-C-D & 2\Delta_A \\ 2 & -A+B-C+D \end{pmatrix}}{A+B+C+D}$
  - $\frac{\begin{pmatrix} \Sigma_{12} & \Delta_\Sigma \\ 1 & -\Sigma_{21} \end{pmatrix}}{\Sigma_{22}}$
  - 

- **Impedanzparameter**:
  - $U_1 = Z_{11} I_1 + Z_{12} I_2$
  - $U_2 = Z_{21} I_2 + Z_{22} I_2$
  - 

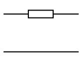
- **Kettenparameter**:
  - Darstellung der ABCD-Parameter
  - $\begin{pmatrix} U_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} U_2 \\ -I_2 \end{pmatrix}$
  - $\frac{\begin{pmatrix} -\Delta_S + S_{11} - S_{22} + 1 & \Delta_S + S_{11} + S_{22} + 1 \\ \Delta_S - S_{11} - S_{22} + 1 & -\Delta_S - S_{11} + S_{22} + 1 \end{pmatrix}}{2S_{21}}$
  - 

- **Transmissionsmatrix**:
  - Multiplikation Vereinfachung
  - $\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$
  - $[\Sigma] = \frac{1 \begin{pmatrix} -\Delta_S & S_{11} \\ -S_{22} & 1 \end{pmatrix}}{S_{21}}$
  - 

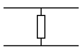
  - 

  - $[\Sigma_g] = [\Sigma_1]\,[\Sigma_2]$

1.1 Leitungen

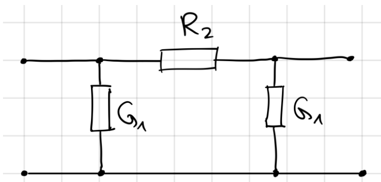


$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & \frac{R}{Z_0} \\ 0 & 1 \end{bmatrix}$



$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ GZ_0 & 1 \end{bmatrix}$

1.1.1 Dämpfungsglied



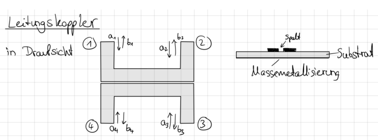
$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{R_2}{Z_0} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{bmatrix}$

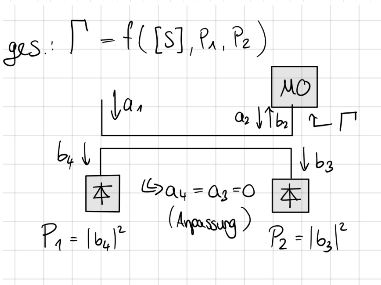
2 Koppler

Taktanregung

- **Gleichtaktanregung**: + Signale gleicher Amplitude und gleicher Phase.
  - Z_e
  - $a_1^+ = \frac{1}{2} a_1, a_4^+ = \frac{1}{2} a_1$
- **Gegentaktanregung**: − Signale gleicher Amplitude und engegengesetzer Phase.
  - Z_d
  - $a_1^- = \frac{1}{2} a_1, a_4^- = -\frac{1}{2} a_1$
- Überlagerung: $a_1^+ + a_1^- = a_1$, $a_4^+ + a_4^- = 0$

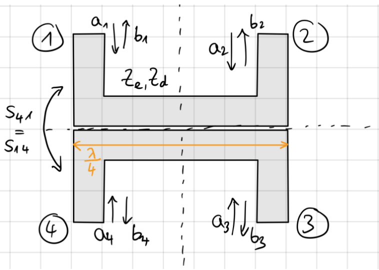
2.1 Leitungskoppler





$|\Gamma|^2 = \frac{P2}{P1} \left| \frac{S_{41}}{S_{32} S_{21}} \right|^2$

Zweifach Symmetrischer Zweitor:



Allseit. Anpassung:
$S_{11} = S_{22} = S_{33} = S_{44} = 0$
Symmetrien und Reziprok:
$S_{41} = S_{14} = S_{32} = S_{23}$
$S_{12} = S_{21} = S_{34} = S_{43}$

Mit Gleich- und Gegentaktanregung:

$b_1 = b_1^+ + b_1^- = S_{11}^+ a_1^+ + S_{11}^- a_1^- = \frac{1}{2}(S_{11}^+ + S_{11}^-)a_1, \rightarrow S_{11}$
$b_2 = b_2^+ + b_2^- = S_{21}^+ a_1^+ + S_{21}^- a_1^- = \frac{1}{2}(S_{21}^+ + S_{21}^-)a_1, \rightarrow S_{21}$
$b_3 = b_2^+ + b_2^- = S_{21}^+ a_4^+ + S_{21}^- a_4^- = \frac{1}{2}(S_{21}^+ - S_{21}^-)a_1, \rightarrow S_{31}$
$b_4 = b_4^+ + b_4^- = S_{11}^+ a_4^+ + S_{11}^- a_4^- = \frac{1}{2}(S_{11}^+ - S_{11}^-)a_1, \rightarrow S_{41}$

Normierung der Widerstände: $z_e = \frac{Z_e}{Z_0}, z_d = \frac{Z_d}{Z_0}$

$$\begin{bmatrix} A^{\pm} & B^{\pm} \\ C^{\pm} & D^{\pm} \end{bmatrix} = \begin{bmatrix} \cos(\beta l) & j z_{\frac{e}{d}} \sin(\beta l) \\ j \frac{1}{z_{\frac{e}{d}}} \sin(\beta l) & \cos(\beta l) \end{bmatrix}$$

$S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} = \frac{j z_e \sin(\beta l) - j \frac{1}{z_e} \sin(\beta l)}{2\cos(\beta l) + j(z_e + \frac{1}{z_e})\sin(\beta l)}$

$S_{11}^- = \frac{j(z_d - \frac{1}{z_d})\sin(\beta l)}{2\cos(\beta l) + j(z_d + \frac{1}{z_d})\sin(\beta l)}$

$S_{21}^+ = \frac{2}{A^+ + B^+ + C^+ + D^+} = \frac{2}{2\cos(\beta l) + j(z_e + \frac{1}{z_e})\sin(\beta l)}$

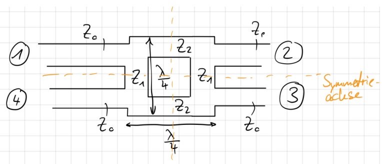
$S_{21}^+ = \frac{2}{A^- + B^- + C^- + D^-} = \frac{2}{2\cos(\beta l) + j(z_d + \frac{1}{z_d})\sin(\beta l)}$

Allseitige anpassung
$S_{11} = 0 \rightarrow S_{11}^+ = -S_{11}^- \rightarrow z_e = \frac{1}{z_d} \leftrightarrow z_e z_d = 1 \rightarrow Z_e Z_d = Z_0^2$
Entkoppelte Tore: $S_{31} = \frac{1}{2}(S_{21}^+ - S_{21}^-) = 0 \rightarrow S_{21}^+ = S_{21}^-$
$S_{21} = S_{21}^+ = \frac{2}{2\cos(\beta l) + j(z_e + \frac{1}{z_e})\sin(\beta l)}$

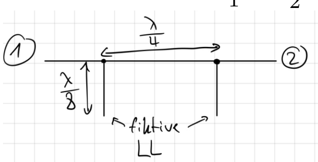
$S_{41} = \frac{1}{2}(S_{11}^+ - S_{11}^-) = S_{11}^+ = \frac{j(z_e - \frac{1}{z_e})\sin(\beta l)}{2\cos(\beta l) + j(z_e + \frac{1}{z_e})\sin(\beta l)}$

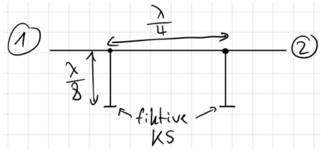
$\angle(S_{21}, S_{41}) = 90°$
Mehrstufig wenn höhere Breitband.

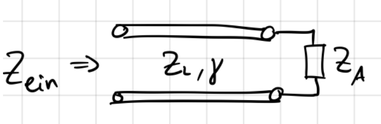
2.2 Ringkoppler



- 3dB-Kopplung: $1 \rightarrow 2$ und $1 \rightarrow 3$
- Entkopplung: 1 und 4
- Einspeisung an 1
- $Z_1 =?$, $Z_2 =?$

- **Gleichtakt**
  - $+$ an 1 und 4: $a_1^+ = \frac{1}{2}a_1, a_4^+ = \frac{1}{2}a_1$
  - 

- **Gegentakt**
  - $-$ an 1 und 4: $a_1^- = \frac{1}{2}a_1, a_4^- = -\frac{1}{2}a_1$
  - 



- $Z_{ein} = Z_L \frac{Z_A + jZ_L \tan(\beta l)}{Z_L + jZ_A \tan(\beta l)}$
- $\tan(\beta l) = \tan(\frac{2\pi}{\lambda}\frac{\lambda}{8}) = \tan(\frac{\pi}{4}) = 1$
- $+$ $Z_A \rightarrow \infty \Rightarrow Z_{ein,LL}^+ = \lim_{Z_A \rightarrow \infty} Z_{ein} = -jZ_L = -jZ_1$
- $-$ $Z_A \rightarrow 0 \Rightarrow Z_{ein,LL}^- = jZ_L = jZ_1 = j\frac{1}{Y_1}$

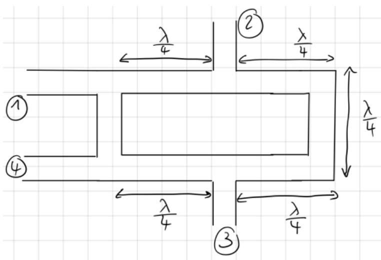
Kettenmatrizen fortsetzung

$$\left[M^{\pm}\right] = \begin{bmatrix} 1 & 0 \\ -jY_1Z_0 & 1 \end{bmatrix} \begin{bmatrix} 0 & j\frac{Z_2}{Z_0} \\ j\frac{Z_0}{Z_2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -jY_1Z_0 & 1 \end{bmatrix} =$$
$$\begin{bmatrix} \pm Z_2Y_1 & j\frac{Z_2}{Z_0} \\ j\frac{Z_0}{Z_2} - jY_1^2 Z_0 Z_2 & \pm Y_1 Z_2 \end{bmatrix} = \begin{bmatrix} A^{\pm} & B^{\pm} \\ C^{\pm} & D^{\pm} \end{bmatrix}$$

Anforderung: Allseitige Anpassung

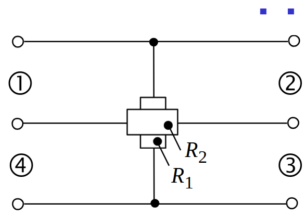
- $S_{11} = \frac{1}{2}(S_{11}^+ + S_{11}^-) \stackrel{!}{=} 0$
- $S_{41} = S_{14} = \frac{1}{2}(S_{11}^+ - S_{11}^-) \stackrel{!}{=} 0$
- $S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} \stackrel{!}{=} 0$
- Da $A^+ = D^+ \Rightarrow B^+ = C^+$
- $\frac{Z_2}{Z_0} = \frac{Z_0}{Z_2} - Y_1^2 Z_2 Z_0$
- $|S_{21}| = |\frac{1}{2}\left(S_{21}^+ + S_{21}^-\right)| = |S_{31}| = |\frac{1}{2}\left(S_{21}^+ - S_{21}^-\right)|$
- $\frac{Z_2}{Z_0} = Y_1 Z_2 \Rightarrow Y_1 = \frac{1}{Z_0}$
- $1 = \frac{Z_0^2}{Z_2^2} - Y_1^2 Z_0^2 \Rightarrow Z_2 = \frac{1}{\sqrt{2}} Z_0$
- $\angle(S_{21}, S_{31}) = 90°$
- 3dB-Kopplung: $1 \rightarrow 2$ und $1 \rightarrow 3$
- Entkopplung: 1 und 4

2.2.1 Rat-Race-Koppler



- Einsp. 1: 3 entkoppelt, $\angle(S_{21}, S_{41}) = 0°$
- Einsp. 3: 1 entkoppelt, $\angle(S_{23}, S_{43}) = 180°$

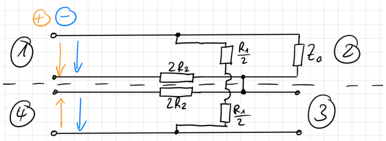
2.3 Resistiver Koppler



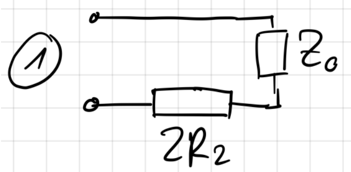
Synthese: All. Anpassung

- $S_{31} = S_{13} = S_{24} = S_{42} = 0$

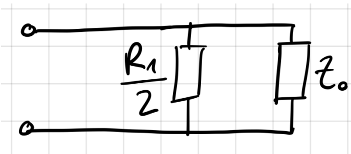
$S_{11} = \frac{Z_{ein} - Z_0}{Z_{ein} + Z_0}$
$+$ und $-$ an 1 und 4.



⊕ $\Rightarrow$ in der Symmetrieeben $\Rightarrow$ LL

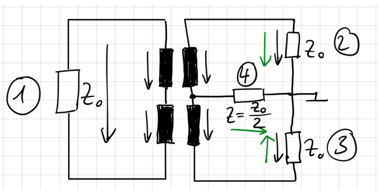
- 
- $S_{11}^{+} = \frac{2R_2+Z_0-Z_0}{2R_2+Z_0+Z_0} = \frac{R_2}{R_2+Z_0}$

(-)

- 
- $S_{11}^{-} = -\frac{Z_0}{R_1+Z_0}$

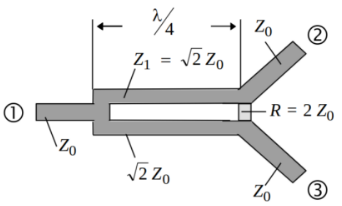
$S_{11} = \frac{1}{2}(S_{11}^{+} + S_{11}^{-}) \stackrel{!}{=} 0 \Rightarrow S_{11}^{+} = -S_{11}^{-}$
$\frac{R_2}{R_2+Z_0} = \frac{Z_0}{R_1+Z_0} \Rightarrow R_1 R_2 = Z_0^2$
$S_{41} = \frac{1}{2}(S_{11}^{-} - S_{11}^{-}) = S_{11}^{+} = \frac{R_2}{R_2+Z_0}$
$S_{21} = \frac{Z_0}{Z_0+R_2}$
Häufige Wahl: $R_1 = R_2 = Z_0 \Rightarrow S_{21} = S_{41} = \frac{1}{2}$
Nullgradkoppler: $\angle(S_{21}, S_{41}) = 0°$

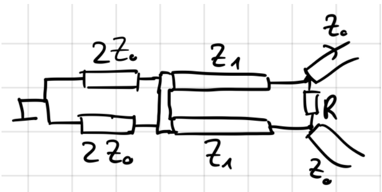
2.4 Transformatorkoppler

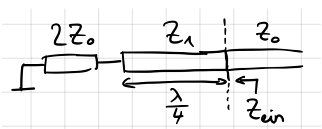
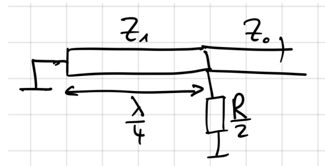


- Transformator sei ideal
- aus Symmetrie sind 1 und 4 entkoppelt
- bei Einspeisung an 1 $\Rightarrow$ Signale an 2 und 3 in Gegenphase
- bei Einspeisung an 4 $\Rightarrow$ Signale an 2 und 3 in Phase
- $\Rightarrow$ Signale an 2 und 3 entkoppelt mit $Z = \frac{Z_0}{2}$

2.5 Nullgradkoppler/Signalteiler

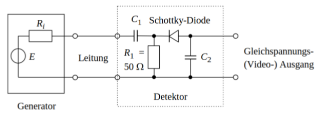




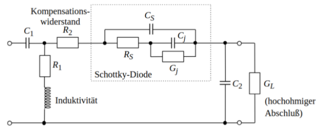
- ⊕ und (-) and 2 und 3
- ⊕ an 2 und 3:
  
  -
  - $Z_{ein} \frac{Z_1}{2Z_0} \stackrel{!}{=} 0 \Rightarrow Z_1 = \sqrt{2} Z_0$
- (-) an 2 und 3:
  
  -
  - $\frac{R}{2} = Z_0 \Rightarrow R = 2Z_0$

3 Messung

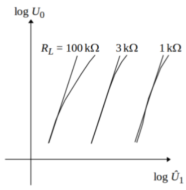
3.1 Breitbanddetektoren



Detektor mit Zwangsanpassungswiderstand R_1



Nicht Lineare

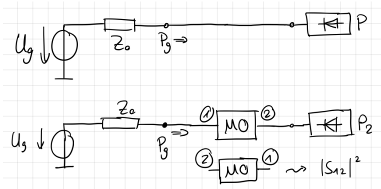


Effekte(Verständnis)

Bild 1.27: Detektorausgangsspannung U_0 als Funktion des HF-Pegels $\hat{U}_1$ für verschiedene Lastwiderstände R_L

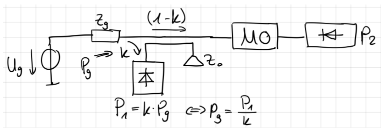
3.2 Skalare Zweitormessung

3.2.1 Substitionsverfahren



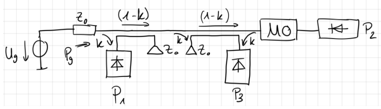
$P_2 = |S_{21}|^2 P_g \Leftrightarrow |S_{21}|^2 = \frac{P_2}{P_1}$

3.2.2 Leitungskoppler



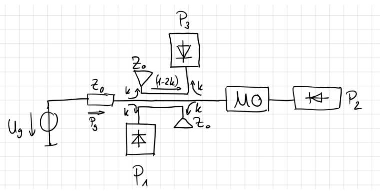
$P_1 = KP_g$

$|S_{21}|^2 = \frac{K}{(1-K)} \frac{P_2}{P_1}$



- $P_2 = |S_{21}|^2 (1-k)^2 P_g$
- $P_1 = kP_g$
- $P_3 = |S_{11}|^2 (1-k)^2 k P_g$
- $|S_{21}|^2 = \frac{P_2}{P_1} \frac{k}{(1-k)^2}$
- $|S_{11}|^2 = \frac{P_3}{P_1} \frac{1}{(1-k)^2}$

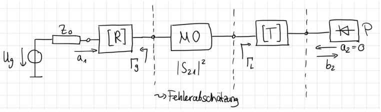
3.2.3

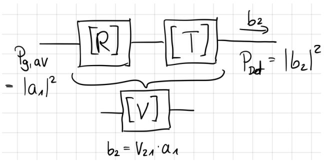
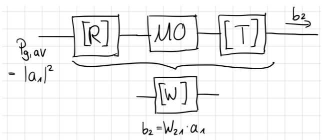


- $P_1 = kP_g$
- $P_2 = |S_{21}|^2(1-2k)P_g$
- $P_3 = k|S_{11}|^2(1-2k)P_g$
- $|S_{21}|^2 = \frac{P_2}{P_1}\frac{k}{1-2k}$
- $|S_{11}|^2 = \frac{P_3}{P_1}\frac{1}{1-2k}$

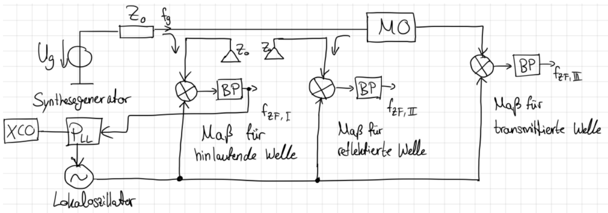
3.2.4 Fehler bei Transmissionmessung

Annahme: Fehlanpassung am Generator und am Leistungsdetektor → Fehlermodell



- $P_{Det} = |b_2|^2$
- Ohne MO:
  - 
  - $P_{g_{ver}} = |a_1|^2$
  - $b_2 = V_{21}a_1$
  - $P_{Det} = |b_1|^2 = |V_{21}|^2|a_1|^2$
  - $R_{22} = \Gamma_g$
  - $T_{11} = \Gamma_L$
  - Zu Transmissionmatrizen
  - $[\Sigma_V] = [\Sigma_R]\,[\Sigma_T] = \frac{1}{R_21}\begin{bmatrix}-\Delta_R & R_{11}\\ -R_{22} & 1\end{bmatrix}\frac{1}{T_21}\begin{bmatrix}-\Delta_T & T_{11}\\ -T_{22} & 1\end{bmatrix}$
  - $V_{21} = \frac{1}{\Sigma_{V_{22}}} = \frac{R_21T_21}{1-R_22T_11}$
  - $P_{Det} = \left|\frac{R_{21}T_{21}}{1-R_{22}T_{11}}\right|^2 P_{g_{ver}}$
  - Annahme: Fehlerzweitoren verlustlos:
    * $|R_{21}|^2 = |R_{12}|^2 = 1-|R_22|^2$
    * $|T_{21}|^2 = 1-|T_11|^2$
  - $P_{Det} = \frac{(1-|\Gamma_g|^2)(1-|\Gamma_L|^2)}{|1-\Gamma_g\Gamma_L|^2}P_{g_{ver}}$
- Mit MO:
  - 
  - $W_{21} = \frac{1}{\Sigma_{W_{22}}}$
  - $\Sigma_W = \frac{1}{R_{21}S_{21}T_{21}}\begin{bmatrix}-\Delta_R & R_{11}\\ -R_{22} & 1\end{bmatrix}\begin{bmatrix}-\Delta_S & S_{11}\\ -S_{22} & 1\end{bmatrix}\begin{bmatrix}-\Delta_T & T_{11}\\ -T_{22} & 1\end{bmatrix}$
  - $W_{21} = \frac{R_{21}S_{21}T_{21}}{1-R_{22}S_{11}-S_{22}T_{11}+T_{11}R_{22}\Delta_S}$
  - $\left|\hat{S}_{21}\right|^2 = \frac{|W_{21}|^2}{|V_{21}|^2} = |S_{21}|^2\frac{|1-\Gamma_g\Gamma_L|^2}{|(1-\Gamma_gS_11)(1-\Gamma_LS_22)-S_{12}S_{21}\Gamma_g\Gamma_L|^2}$
- Beispiel für Fehlerschätzung:
  - MO: $|S_{12}| = |S_{21}| = \frac{1}{2}$, $|S_{11}| = |S_{22}| = 0.1$
  - Messsystem: $|\Gamma_g| = 0.1$, $|\Gamma_L| = 0.1$
  - $\left|\hat{S}_{21}\right|^2 = \frac{|1\pm0.01|^2}{|(1\mp0.01)^2\mp0.0025|^2}|S_{21}|^2$
  - $\left|\hat{S}_{21}\right|^2 \approx \frac{1\pm0.02}{(1\mp0.02\mp0.0025)^2}|S_{21}|^2$
  - $\left|\hat{S}_{21}\right|^2 \approx \frac{1\pm0.02}{1\mp0.045}|S_{21}|^2 \approx (1\pm0.06)|S_{21}|^2 \hat{=} 0.27dB$

3.3 Vektorielle Netzwerkanalyse



- Maß für fortlaufende Welle $f_{ZF,I}$
- Maß für reflektierte Welle $f_{ZF,II}$
- Maß für transmitierte Welle $f_{ZF,III}$
-